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De novo synthesis of benzofurans *via* trifluoroacetic acid catalyzed cyclization/oxidative aromatization cascade reaction of 2-hydroxy-1,4-diones†

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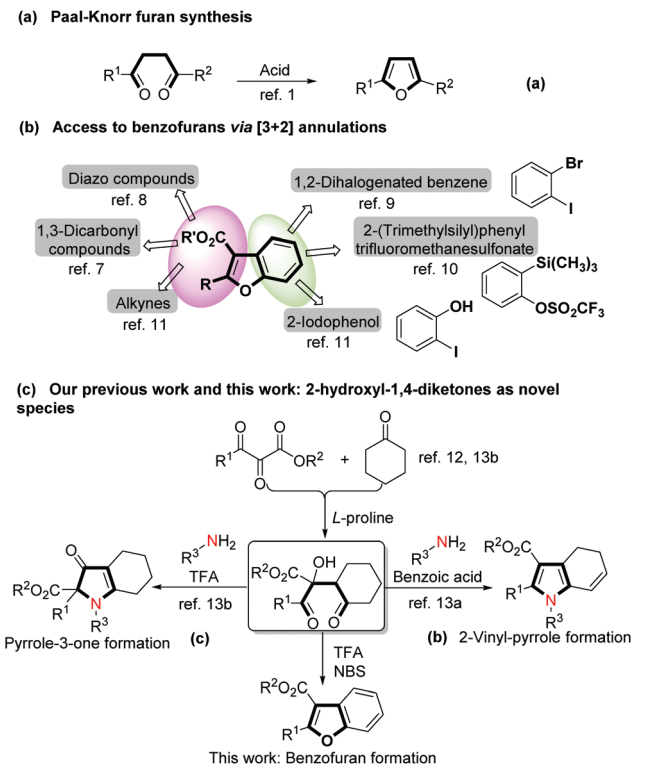
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A facile, one-pot method for the *de novo* synthesis of benzofurans from 2-hydroxy-1,4-diones is described. Using trifluoroacetic acid (TFA) as the catalyst and *N*-bromobutanimide (NBS) as the oxidant, 2-hydroxy-1,4-diones underwent a cyclization/oxidative aromatization cascade reaction to afford a variety of benzofuran derivatives in moderate to good yields. This is a practically useful method for the synthesis of benzofuran derivatives starting from 2-hydroxy-1,4-diones which is also a supplement of the Paal–Knorr furan synthesis. A preliminary mechanism study was undertaken to support the proposed mechanism during which a novel 1,6-conjugate addition reaction was revealed.

Introduction

Paal–Knorr furan synthesis, developed by Paal and Knorr in 1884,[1] is one of the most widely used methods (Scheme 1a) to generate furan derivatives which are prevalent in natural products, biologically active compounds, and synthetic pharmaceuticals and materials.[2] Along with the increasing methods to access 1,4-diones substrates,[3] Paal–Knorr furan synthesis has received a lot of attention and found many applications.[4] Research in this field mainly focused on the synthesis of furan derivatives; nevertheless, benzofuran formation from 1,4-diones has rarely been realized until now.[5]

Benzofurans are important skeletons that widely exist in natural products, biologically active compounds and drugs.[6] Generally, they were synthesized *via* formal [3 + 2] annulations (Scheme 1b) from 1,3-dicarbonyl derivatives (including 1,3-dicarbonyl compounds[7] and diazo compounds[8]) and benzene ring precursors (1,2-dihalogenated benzene[9] and 2-(trimethylsilyl)phenyl trifluoromethanesulfonate[10]). Moreover, they can be obtained from intermolecular or intramolecular annulation of a phenol and an alkyne.[11] Since the benzene ring mainly derives from aromatic reagents and mostly requires transition-metal catalysts, it is highly demanded to develop new methods that use aliphatic compounds as reactants in which transition-metal-free conditions could be used.



Scheme 1 General routes toward furans and benzofurans.

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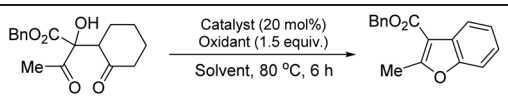
‡These authors contribute equally to this work.

Recently, we have developed L-proline catalyzed aldol reactions between 2,3-diketoesters and cyclohexanones to afford a variety of novel 2-hydroxy-1,4-diones (Scheme 1c).[12,13b] Further research studies based on these 2-hydroxy-1,4-diones demonstrated their uses in the synthesis of 2-vinyl-pyrroles and pyrrole-3-ones.[13] Combined with our knowledge of 2-vinyl-pyrrole formation, 4-hydroxybenzofuran formation[14] and Paal–Knorr furan synthesis, we envisioned that an appropriate oxidant might allow 2-hydroxy-1,4-diones to undergo a cyclization/oxidation aromatization cascade reaction to form benzofurans. Herein, we wish to describe a novel, cascade *de novo* reaction to access benzofurans from 2-hydroxy-1,4-diones. It is worth mentioning that the hydroxyl group played a crucial role in the reaction which dominants the dehydration to form the key C=C bond during the cascade reaction process.

Results and discussion

At the outset, we chose **2a** as the model substrate to explore suitable reaction conditions. Since DMSO is commonly used in the oxidation process,[15] we decided to use DMSO as both the oxidant and the solvent. Acetic acid was chosen as the catalyst and the reaction was run at room temperature for 6 h. However, no desired product **3a** was detected (Table 1, entry 1). When the reaction temperature was increased to 80 °C, we were delighted to obtain **3a** in 31% yield. With this inspiring result, we further optimised the reaction using various catalysts, including CF_3SO_3H, $BF_3{\cdot}Et_2O$, *t*-BuCOOH and CF_3COOH. CF_3COOH was shown to be the most favourable catalyst (Table 1, entries 2–6) which gave **3a** in 42% yield. However, it was disappointing to find out that on using DMSO as the oxidant, **3a** was always contaminated by inseparable by-products (column chromatography) which reduced the practical applications of the reaction. Then the optimization strategy was changed to examine various oxidants. DCE was used as the solvent. Oxidants including BQ, MnO_2, $K_2S_2O_8$, *m*-CPBA, DDQ, DIB and NBS were screened. The results showed that using BQ, MnO_2, $K_2S_2O_8$ and *m*-CPBA gave no desired product (Table 1, entries 7–10). DDQ and DIB gave **3a** in 30% and 25% yields, respectively (Table 1, entries 11 and 12). The yield of **3a** was greatly improved by using NBS as the oxidant (Table 1, entry 13). The reaction didn't occur in the absence of the catalyst or when the reaction temperature was lowered to room temperature, indicating the necessity of the catalyst and external heat (Table 1, entries 14 and 15).

Table 1 Optimization of the reaction conditions[a]



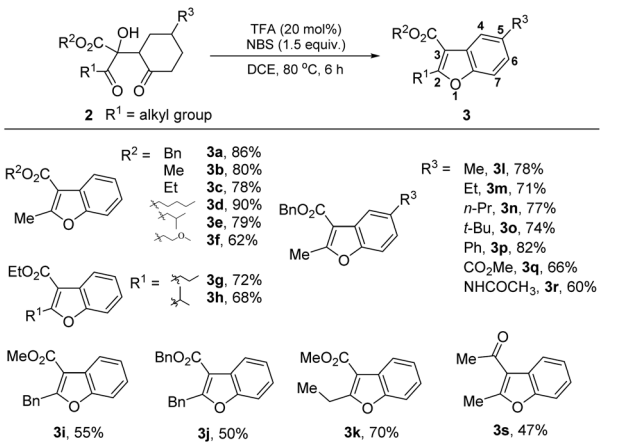
2a **3a**

Entry	Catalyst	Solvent	Oxidant	Temp. (°C)	Yield[b] (%)
1	CH_3COOH	DMSO	—	25	0
2	CH_3COOH	DMSO	—	80	31
3	CF_3SO_3H	DMSO	—	80	15
4	$BF_3{\cdot}Et_2O$	DMSO	—	80	<10
5	*t*-BuCOOH	DMSO	—	80	Trace
6	CF_3COOH	DMSO	—	80	42[c]
7	CF_3COOH	DCE	BQ	80	Trace
8	CF_3COOH	DCE	MnO_2	80	Trace
9	CF_3COOH	DCE	$K_2S_2O_8$	80	Trace
10	CF_3COOH	DCE	*m*-CPBA	80	Trace
11	CF_3COOH	DCE	DDQ	80	30
12	CF_3COOH	DCE	DIB	80	25
13	**CF_3COOH**	**DCE**	**NBS**	**80**	**86**
14	—	DCE	NBS	80	0
15	CF_3COOH	DCE	NBS	25	0

[a] **2a** (0.2 mmol), catalyst (0.2 equiv.), solvent (2.0 mL), and oxidant (1.5 equiv.), at the indicated temperature for 6 h. [b] Yields refer to isolated yields. [c] Contaminated by inseparable by-products.

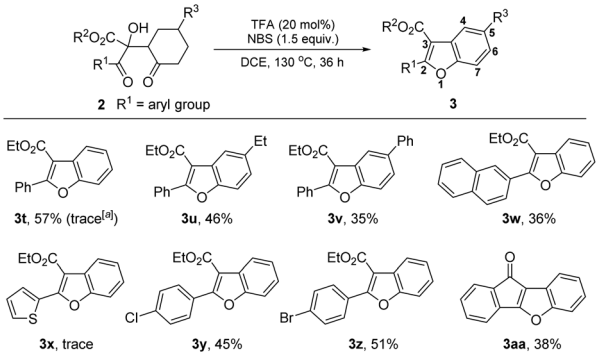
With the optimum conditions in hand, the scope and limitations of this transformation were then examined. Reactants with various ester groups gave **3** in good yields (Scheme 2, **3a–3f**). Substrates with various alkyl groups (R^1 group), such as ethyl, *n*-propyl, i-propyl and benzyl groups, also gave substituted benzofurans in moderate to good yields (Scheme 2, **3g–3k**). Subsequently, substrates with different R^3 groups were tested. The results indicated that alkyl, aryl, ester and amide groups were all tolerated and gave **3** in good yields (Scheme 2, **3l–3r**). When the ester group in the substrate was replaced by a ketone group, the reaction occurred smoothly as well and **3s** was obtained in 47% yield.

Further exploration of the substrates with an aryl group at the ketone unit (R^1) failed by using the previous optimized conditions because **3t** was obtained in trace amounts (Scheme 3, **3t**). We speculated that the large steric hindrance of the phenyl ring might make the intramolecular annulation much more difficult. Therefore, we increased the reaction temperature hoping to improve the yield of **3t**. Fortunately, **3t** was isolated in 57% yield when performing the reaction at 130 °C for 36 h. Two representative substrates, one with an alkyl substituent (R^3 = Et) and the other with an aryl substituent (R^3 = Ph) were studied, and the corresponding products



Scheme 2 Scope of benzofuran formation (R^1 = alkyl group). **2** (0.2 mmol), TFA (0.2 equiv.), DCE (2.0 mL), and NBS (1.5 equiv.), at 80 °C for 6 h; yields refer to isolated yields.

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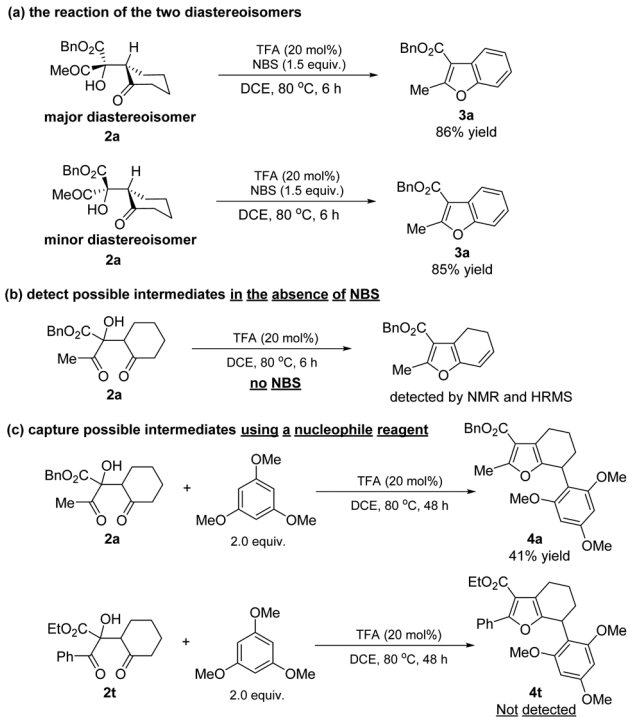


Scheme 3 Scope of benzofuran formation (R^1 = aryl group). **2** (0.2 mmol), TFA (0.2 equiv.), DCE (2.0 mL), and NBS (1.5 equiv.), at 130 °C for 36 h; yields refer to isolated yields. [a] The reaction was performed at 80 °C for 6 h.

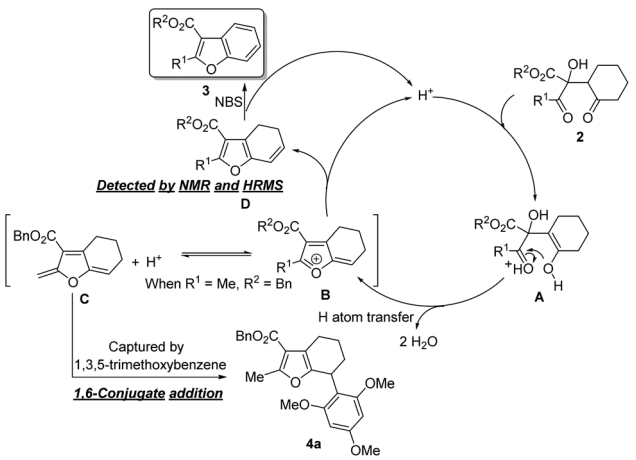
were obtained in 46% and 35% yields, respectively (Scheme 3, **3u** and **3v**). Other substrates with aryl groups at the ketone unit were investigated. Naphthyl ring and *p*-halophenyl groups (Cl and Br) were well tolerated and the desired products were isolated in moderate yields (Scheme 3, **3w**, **3y**, and **3z**). Reactants with the 2-thiophene group only gave trace amounts of the product (Scheme 3, **3x**). We were pleased to find that the substrate derived from ninhydrin also reacted well, affording benzofuroindenone **3aa** in 38% yield.

In order to understand the mechanism of this transformation, several experiments were undertaken. Since reactants **2** were used as the mixture of diastereoisomers, we were curious about the reactivity of each diastereoisomer. We first chose the two diastereoisomers of **2a** as a representative starting material to undergo the reaction. As observed, similar yields of **3a** were obtained, which indicates the similar reactivity of each diastereoisomer (Scheme 4(a)). Next, in order to detect possible intermediates, we performed the reaction without NBS. Fortunately, we were able to detect the 4,5-dihydrobenzofuran derivative by both ^{1}H NMR and HRMS (Scheme 4(b); for details, see the ESI†). Finally, we tried to capture possible reaction intermediates by using a highly reactive nucleophile. By using two equivalents of 1,3,5-trimethoxybenzene,[16] we were delighted to obtain **4a** in 41% yield which was proposed to form *via* the 1,6-conjugate addition of a conjugated alkene intermediate (Scheme 4c and 5).[17] When **2t**, a representative substrate with an aryl group attached to the carbonyl group, was used, the corresponding product **4t** was not formed.

Based on the above mechanistic studies, a reasonable reaction mechanism was proposed (Scheme 5). Initially, acid catalyzed intramolecular cyclization of **2** occurred, after which a dehydration reaction and hydrogen atom transfer occurred to afford the key intermediate **B**. When R^1 is an alkyl group, **B** can release a H^+ to form intermediate **C**, which can be successfully captured by 1,3,5-trimethoxylbenzene *via* 1,6-conjugate addition. To some extent, the formation of **4a** from intermediate **B** can be excluded because when substrate **2t** (R^1 = phenyl) is used, the 1,6-conjugate addition reaction can't occur. In the absence of 1,3,5-trimethoxylbenzene, **B** will then convert to



Scheme 4 Mechanistic study.



Scheme 5 Proposed mechanism.

intermediate **D** along with the elimination of H^+ which enters the next catalytic cycle. Finally, **D** was oxidized by NBS to form benzofuran derivatives **3**. It is worth noting that during the oxidation step, one equivalent of HBr was formed which could also promote the cyclization reaction.

Experimental

General procedure for the synthesis of benzofuran derivatives

Procedure for the synthesis of 3a–3r. A Schlenk tube with a magnetic stir bar was charged with 2-hydroxy-1,4-diones **2**

(0.20 mmol), trifluoroacetic acid (0.04 mmol, 0.2 equiv.), 1,2-dichloroethane (DCE, 2.0 mL) and *N*-bromobutanimide (NBS, 0.30 mmol, 1.5 equiv.). The reaction mixture was then heated to 80 °C and stirred for 6 hours. It was then allowed to cool to ambient temperature and all of the volatiles were removed under vacuum; the crude product was purified by flash chromatography and eluted with petroleum ether/ethyl acetate to provide substituted benzofuran **3**.

Procedure for the synthesis of 3t–3aa. A Schlenk tube with a magnetic stir bar was charged with 2-hydroxy-1,4-diones **2** (0.20 mmol), trifluoroacetic acid (0.04 mmol, 0.2 equiv.), 1,2-dichloroethane (DCE, 2.0 mL) and *N*-bromobutanimide (NBS, 0.30 mmol, 1.5 equiv.). The reaction mixture was then heated to 130 °C and stirred for 36 hours. It was then allowed to cool to ambient temperature and all of the volatiles were removed under vacuum; the crude product was purified by flash chromatography and eluted with petroleum ether/ethyl acetate to provide substituted benzofuran **3**.

Conclusions

In summary, we have developed an efficient one-pot protocol for the synthesis of benzofuran derivatives starting from 2-hydroxy-1,4-diones. The reaction features easily available starting materials, simple reaction conditions and a broad substrate scope. Studies on the further applications of these novel 2-hydroxy-1,4-diones in the synthesis of highly functionalized furans and 2-methylene-2,3-dihydrofuran derivatives are underway in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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